

Arij Asad

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Research Profile

Final-year PhD researcher in applied mathematics at the University of Bristol, studying buoyancy-driven mixing and plume-generated stratification in confined environments, with applications to nuclear-waste storage. Methods include Planar Laser-Induced Fluorescence (PLIF), mathematical modelling, similarity analysis and finite-element simulation. **Research interests:** buoyant plumes, mixing and stratification; experimental fluid dynamics; molecular processes in fluids; and computational fluid dynamics and finite-element methods.

Education

University of Bristol November 2022 – October 2026 (expected)

PhD research in Applied Mathematics and Fluid Dynamics

Mechanical Engineering; Faculty of Science and Engineering

- Supervisors: Dr Andrew Lawrie and Dr Stuart Thomson.
- Thesis: *Mixing behaviour of plumes.*

University of Cambridge October 2019 – June 2022

BA (Cantab), Mathematics (Mathematical Tripos), Murray Edwards College

- MA (Cantab), conferred April 2026.

Research and Professional Experience

MSc Research Project Support: Pengzhu Shi January 2025 – September 2026

University of Bristol

Provided day-to-day research guidance on *Interaction of twin buoyant plumes in a confined environment*.

Widening Participation Tutor July 2023 – present

Faculty of Science and Engineering, University of Bristol

Delivered university-access sessions for school pupils aged 11–18.

She Talks Science STEP Summer School July 2023 – August 2025

Murray Edwards College, University of Cambridge

- Director, October 2024 – August 2025: directed a four-day residential programme for 50 students; co-led selection from 184 applications; coordinated five tutors, three supervisors and three enrichment speakers.
- Led academic content, timetabling, staff briefing and safeguarding; liaised with Cambridge Mathematics and delivered inequalities teaching and Cambridge-style supervisions.
- Deputy Director, July 2023 – August 2024: contributed to teaching materials, classes, supervisions and residential activities. The 2024 cohort comprised 49 students, of whom 28 subsequently applied to Cambridge.

Summer Research Project July 2022 – August 2022

Centre for Mathematical Sciences, University of Cambridge

The motion of anisotropic carbon nanoparticles in shear flow: a hydrodynamic perspective. Supervisor: Dr Catherine Kamal.

Research Outputs

Manuscripts in preparation

- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *On the mixing behaviour of plumes in a confined environment*.
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a laminar crossflow*.
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a confined circular domain with and without laminar crossflow*.

Research software: PLIF — MATLAB tools for PLIF image processing, density-field extraction and stratification analysis.

Selected Presentations and Posters

Osborne Reynolds Day	2026
<i>On the mixing behaviour of plumes in a confined environment</i>	
British Applied Mathematics Colloquium (BAMC), Norwich	March 2026
<i>On the mixing behaviour of plumes in a confined environment (accepted poster)</i>	
NWS RSO Annual Conference, Manchester	2026
<i>How does hydrogen build up in a closed vault environment? (poster)</i>	
2nd European Fluid Dynamics Conference (EFDC2), Dublin	2025
<i>Mixing behaviour of plumes</i>	
NWS RSO Annual Conference, Bristol	2025
<i>Hydrogen ventilation in a GDF (poster)</i>	
NWS RSO Annual Conference, Sheffield	2024
<i>Mixing behaviour of laminar plumes with and without a ventilating crossflow</i>	
NWS RSO Advanced Manufacturing Webinar (online)	October 2023
<i>Ventilation of hydrogen in UILW vaults (recording)</i>	
NWS RSO Annual Conference, Leeds	2023
<i>Hydrogen ventilation in a geological disposal facility</i>	
Cambridge Mathematics Placements (CMP) Presentation Day	2022
<i>The motion of anisotropic carbon nanoparticles in shear flow: a hydrodynamic perspective</i>	

Examining and Admissions

Marking Supervisor	2026
<i>OCR (Oxford Cambridge and RSA)</i>	
- STEP Mathematics 2 (Mechanics): supervised and quality-assured marking, monitored consistency and provided corrective feedback to markers. - Reviewed script zoning to resolve ambiguous question allocations.	
STEP Marker	2024 – present
<i>OCR (Oxford Cambridge and RSA)</i>	
- STEP Mathematics 3, 2024–2026; STEP Mathematics 2, 2025. - Applied mark schemes and provided feedback on their implementation.	
Mathematics Undergraduate Admissions Interviewer	December 2024
<i>Homerton College, University of Cambridge</i>	
Served on a two-person panel scheduled for 49 online interviews over four days; assessed mathematical problem solving and completed formal admissions reports.	
Assessment Associate	2024
<i>Pearson VUE</i>	
Reviewed mathematics assessment material for technical accuracy and clarity.	
Marker	2024 – 2025
<i>University of Bristol</i>	
- Applied Analysis B and Probability and Statistics, 2025. - Applied Linear Algebra Coursework, Engineering Mathematics and Introduction to Pure Mathematics, 2024.	
Vetter	2019
<i>United Kingdom Mathematics Trust</i>	
Assessed draft UK Mathematical Olympiad for Girls papers for content, scope, difficulty and clarity, and provided feedback.	
Marker	October 2019
<i>United Kingdom Mathematics Trust</i>	
UK Mathematical Olympiad for Girls.	

Teaching Experience

Supervisor

Michaelmas 2024 – present

University of Cambridge, across ten colleges

Part II Fluid Dynamics II.

- Delivered paired supervisions, marked written work and led revision teaching for 29 students across ten colleges over two academic years (11 in 2024–2025; 18 in 2025–2026).
- Reappointed for ten pairs in 2026–2027; provided academic references and support with Part III and postgraduate progression.

Lead Demonstrator

September 2024 – March 2025

University of Bristol

Wrote and delivered weekly tutorials.

- Probability and Statistics, January 2025 – March 2025.
- Introduction to Pure Mathematics, September 2024 – December 2024.

Graduate Teacher / Demonstrator

September 2023 – present

University of Bristol

- Engineering Mathematics, September 2024 – January 2026.
- Modelling in Applied Mathematics, September 2024 – December 2024.
- Applied Linear Algebra, March 2024 – April 2024.
- Continuum Mechanics, January 2024 – May 2024.

Private Mathematics Tutor

October 2019 – present

Arij Asad Maths; primarily online, with occasional in-person teaching

- A-level Mathematics, Further Mathematics, Physics and Computer Science, October 2019 – present.
- STEP and TMUA preparation, October 2020 – present; MAT preparation, 2020 – 2025.
- Delivered structured university mock interviews and written feedback; independent contractor for Blue Education since November 2019.

Outreach and Service

Specialist Mathematics Tutor

July 2026

Dipont Education Summer Camp, Wuxi

Outreach Speaker

July 2024 – July 2026

University of Bristol

Delivered research talks for widening-participation programmes. Insight into Bristol Summer School, July 2024, July 2025 and July 2026; Next Step Bristol, November 2025; Virtual Summer School, May 2025 and May 2026; and Sutton Trust Summer School, July 2024.

Summer School Teaching

2019 – 2022

University of Cambridge

Delivered classes and supervisions and ran evening activities for the Murray Edwards STEP Summer School, 2019 – 2022; the Sutton Trust Summer School, 2021 – 2022; and Intro to Maths at Uni, July 2021.

Speaker

2019

Murray Edwards College Maths Circle, Cambridge

Sphere packing.

Panellist

- COMPOS interview webinar, Blue Education, August 2026.
- Blue Education Oxbridge Interview Seminar, October 2025.
- Mathematics Open Day, University of Cambridge, 2022.
- Sutton Trust Experience Cambridge Day, 2020 – 2022.

Technical Skills and Languages

- Programming and technical computing: MATLAB, Python and C; scientific typesetting in $\text{\LaTeX}$.
- Experimental methods: PLIF image acquisition and MATLAB-based image processing.
- Numerical methods: FEniCS finite-element simulation of incompressible flow and advection–diffusion.
- High-performance computing: parallel simulations on Bristol’s BlueCrystal and BluePebble systems.
- Urdu: reading and conversational proficiency.